

1. Administrative & Study Identification

Full Study Title: A Study to Learn if Multiple Doses of the Study Medicine Called Carbamazepine Changes How the Body Processes the Other Study Medicine PF-07248144 **Official Regulatory Title:** A PHASE 1, OPEN-LABEL, FIXED SEQUENCE, 2-PERIOD STUDY IN HEALTHY PARTICIPANTS TO INVESTIGATE THE EFFECT OF CARBAMAZEPINE ON PF-07248144 PHARMACOKINETICS **Short Title/Acronym:** PF-07248144 & Carbamazepine DDI Study **Protocol Number:** C4551007 **NCT Number:** NCT07198035 **Sponsor Name:** Pfizer **Investigational Product:** PF-07248144 and Carbamazepine **Phase of Development:** Phase I **Study Status:** Recruiting **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the effect of multiple doses of carbamazepine on the pharmacokinetics (PK) of PF-07248144 in healthy participants. **Secondary Objectives:** To evaluate the safety, tolerability, and how PF-07248144 is changed and removed from the body after taking it alone compared to when co-administered with carbamazepine.

2.2 Background & Rationale Carbamazepine is known to change how the body processes some medications. This study is being conducted to understand the drug-drug interaction (DDI) and characterize the pharmacokinetics (how the drug is changed and eliminated from the body) of PF-07248144 when administered alone and when taken with carbamazepine.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Self-controlled) **Blinding:** Open-label **Randomization:** N/A (Fixed sequence, 2-period study)

3.2 Planned Enrollment & Duration Target \$N\$: 12 participants **Number of Sites:** Not Specified (Phase 1 Clinical Research Unit) **Estimated Study Start:** N/A **Estimated Primary Completion:** N/A

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Females of nonchildbearing potential and males, inclusive, at screening who are able to provide consent and are overtly healthy as determined by medical

evaluation including medical history, physical examination, laboratory tests, vital signs, and 12-lead electrocardiogram (ECG)s. 2. Body mass index (BMI) of 16-32 kg/m²; and a total body weight >45 kg (99 lb).

4.2 Exclusion Criteria 1. Participants shown to carry or be positive for human leukocyte antigen (HLA)-B1502 and/or HLA-A3101 (genotyping alleles/markers related with carbamazepine-associated Stevens-Johnson syndrome (SJS) or toxic epidermal necrolysis [TEN]). 2. Any medical or psychiatric condition including any active suicidal ideation in the past year or suicidal behavior in the past 5 years or laboratory abnormality that may increase the risk of study participation or, in the investigator's judgment, make the participant inappropriate for the study. 3. Use of prescription or nonprescription drugs and dietary and herbal supplements within 14 days or 5 half-lives (whichever is longer) prior to the first dose of study intervention. 4. Additionally, current use or anticipated need for food or drugs that are known moderate or strong inducers or inhibitors of CYP2C9 or CYP3A4, including their administration within 14 days or 5 half-lives of the strong inducers or inhibitors of CYP2C9 or CYP3A4, whichever is longer, prior to first dose of study intervention, during the treatment period, and within 10 days after the last dose of PF-07248144. 5. Pregnant female participants; breastfeeding female participants; individual(s) of childbearing potential (IOCBPs); fertile male participants who are unwilling or unable to use highly effective methods of contraception as outlined in this protocol for the duration of the study and for at least 103 days after the last dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: *Period 1:* PF-07248144 administered alone. *Period 2:* Multiple doses of Carbamazepine administered alongside PF-07248144. **Route of Administration:** Oral **Duration of Treatment:** 2-period fixed sequence, with blood sampling up to 8 days following each PF-07248144 dose.

5.2 Concomitant & Prohibited Therapy Permitted: Standardized meals and hydration according to the clinical research unit protocol. **Prohibited:** Use of food or drugs that are known moderate or strong inducers or inhibitors of CYP2C9 or CYP3A4 within 14 days or 5 half-lives (whichever is longer) prior to the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -1 Informed Consent, Medical History, ECG, Labs, Genotyping
Period 1			Day 1 to 8

| PF-07248144 dosing, dense PK blood sampling, AE monitoring | |
Period 2 | Day 1 to 8 | Carbamazepine multiple doses +
PF-07248144, dense PK blood sampling | | **End of Study** | Follow-up
| Final Vital Signs, Safety Labs, Exit Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Maximum Observed Plasma Concentration (C_{max}) Period 1 2. C_{max} Period 2 3. Area Under the Curve From Time Zero to Extrapolated Infinite Time (AUC_{inf}) Period 1 *(These metrics determine how much of the drug is in the blood at different times up to 8 days post-dose).* **Measurement Method:** Sequential central laboratory analysis of multiple blood samples.

7.2 Secondary & Exploratory Endpoints 1. Number of Treatment-Emergent Treatment-Related AEs by Severity 2. Number of Participants With Laboratory Test Abnormalities 3. Number of Participants With Clinically Significant Physical Examination Abnormalities

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical evaluation, vital signs tracking, ECGs, and clinical laboratory tests during the confinement period. **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor/Regulatory bodies. **Data Monitoring Committee (DMC):** Internal Safety Review by Pfizer Phase 1 clinical team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all participants receiving \$\\ge 1\$ dose) and PK Concentration Population. **Sample Size Justification:** Sample size adequately powered for a standard Phase 1 DDI study design to estimate the ratio of geometric means for PK parameters. **Handling of Missing Data:** Standard non-compartmental analysis (NCA) procedures; values below the limit of quantification will be treated according to standard PK imputation rules.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous onsite monitoring at the Phase 1 Clinical Research Unit. **Audit Trail:** eCRF locking, source data verification, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: C4551007 **Registry Number:** NCT07198035 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn if Multiple Doses of the Study Medicine Called Carbamazepine Changes How the Body Processes the Other Study Medicine PF-07248144 **Official Regulatory Title:** A PHASE 1, OPEN-LABEL, FIXED SEQUENCE, 2-PERIOD STUDY IN HEALTHY PARTICIPANTS TO INVESTIGATE THE EFFECT OF CARBAMAZEPINE ON PF-07248144 PHARMACOKINETICS

12. Clinical Framework (Pharmacokinetics Specific)

Primary Condition: Healthy Adults **Diagnosis Threshold:** Overtly healthy (BMI 16-32 kg/m2, Weight >45 kg) **Medical Methodology:** Drug-Drug Interaction (DDI) Evaluation **Optimization Target:** Pharmacokinetic profile characterization

13. Study Procedures & Methodology

Management Pathway: Phase 1 Clinical Unit Confinement and Follow-up **Education Protocol (HCP):** Protocol specific Phase 1 investigator training **Education Protocol (Patient):** Informed consent and unit confinement guidelines **Quality Audit Mechanism:** Source data verification and protocol adherence tracking

14. Observation & Follow-Up Schedule

Total Duration: 2 study periods + up to 8 days post-dose blood sampling per period **Onsite Visit Cadence:** Continuous confinement during dosing and PK dense-sampling phases **Remote Monitoring Frequency:** N/A (Inpatient Phase 1 setting) **Checklist Review Interval:** Daily during the inpatient confinement periods

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				